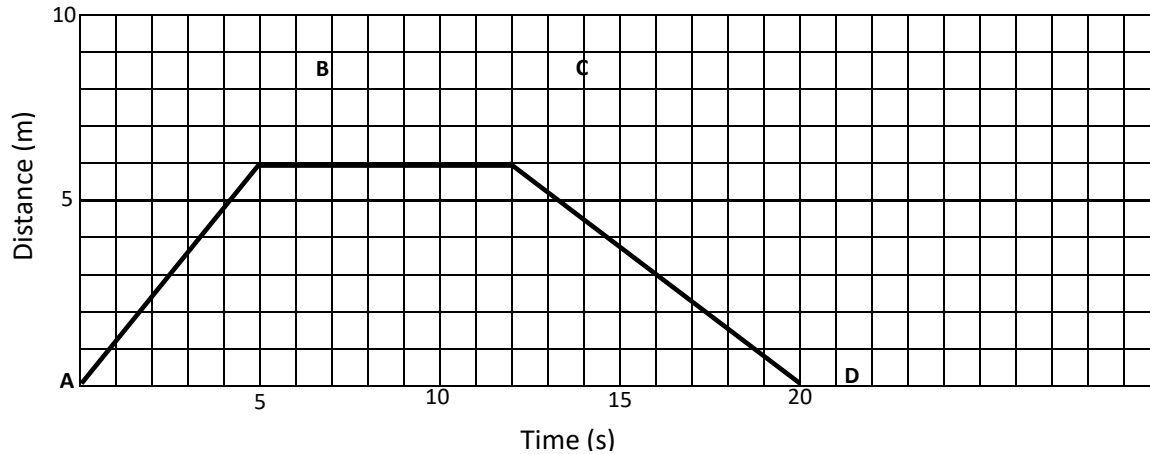


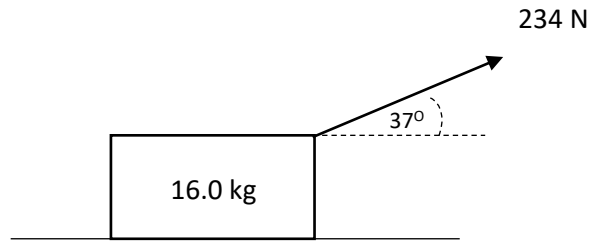
1. A lizard climbs the tree, takes some time to watch flying insects, and then moves back to where he comes from. The graph below shows the lizard's movement plotted in y (distance) and in x (time) axes.



Observe the graph and answer the following questions.

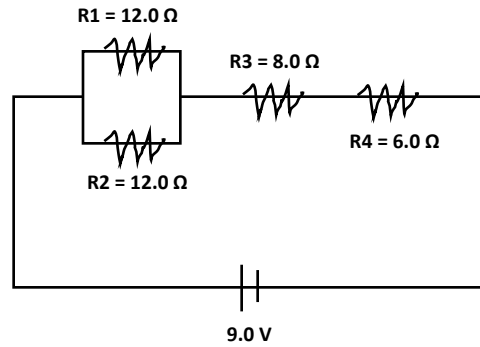
- a) The part of the graph that shows that the lizard is not moving. _____
- b) The part of the graph that shows that the lizard is returning to its initial position. _____
- c) Calculate the speed of the lizard during its journey.
- d) Calculate the average velocity of the lizard during its journey.
- e) If the lizard jumps from the top of the tree with an initial velocity of **0.85 m/s** and able to reach the bottom after **1.4 s**, find the height of the tree.

2. A chain is attached to a big rock of mass **16.0 kg** as shown in the figure below. If the chain is pulled with a force of **234 N** at an angle of **37°** with the horizontal and the coefficient of kinetic friction between the rock and the ground is **0.223**, calculate



- a) The vertical component of the applied force
- b) The horizontal component of the applied force
- c) The normal force on the rock (use $g = 9.81 \text{ m/s}^2$)

3. Analyze the circuit below and answer the following questions.



a) Calculate the equivalent resistance of R1 and R2

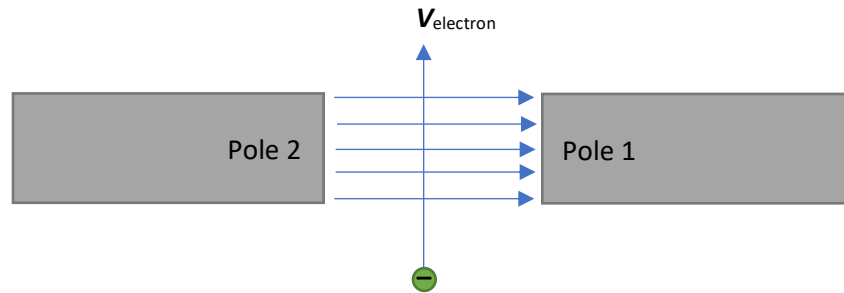
b) Calculate the equivalent resistance of the circuit

c) Calculate the total current of the circuit

d) Calculate the potential difference across R4

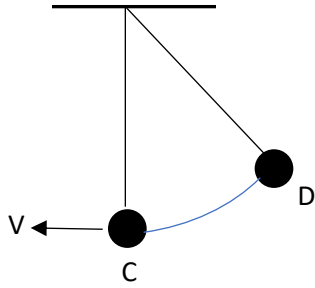
e) Find the total charge in R4 when its current passes through it in **18 s**.

4. An electron ($q = -1.6 \times 10^{-19} \text{ C}$) experiences a force of $5.5 \times 10^{-17} \text{ N}$ as it moves with a velocity of $4.7 \times 10^4 \text{ m/s}$ across the two magnets as shown in the below figure.



- Find the magnitude of the magnetic field.
- What is the correct direction of its magnetic force?
- Label the correct polarity of the magnets.

5. The figure below is a simple pendulum with a 1.0-kg mass attached to it. Its maximum displacement is found at point D and its maximum speed of 0.28 m/s is found at point C. If the pendulum completes 5 cycles in 6.0 s, answer the following questions.



- Calculate the frequency of this device
- Find its period of vibration
- Find the length of the pendulum
- If the pendulum swings a total angle of 40° , find its amplitude
- Calculate the kinetic energy at the lowest position of the pendulum